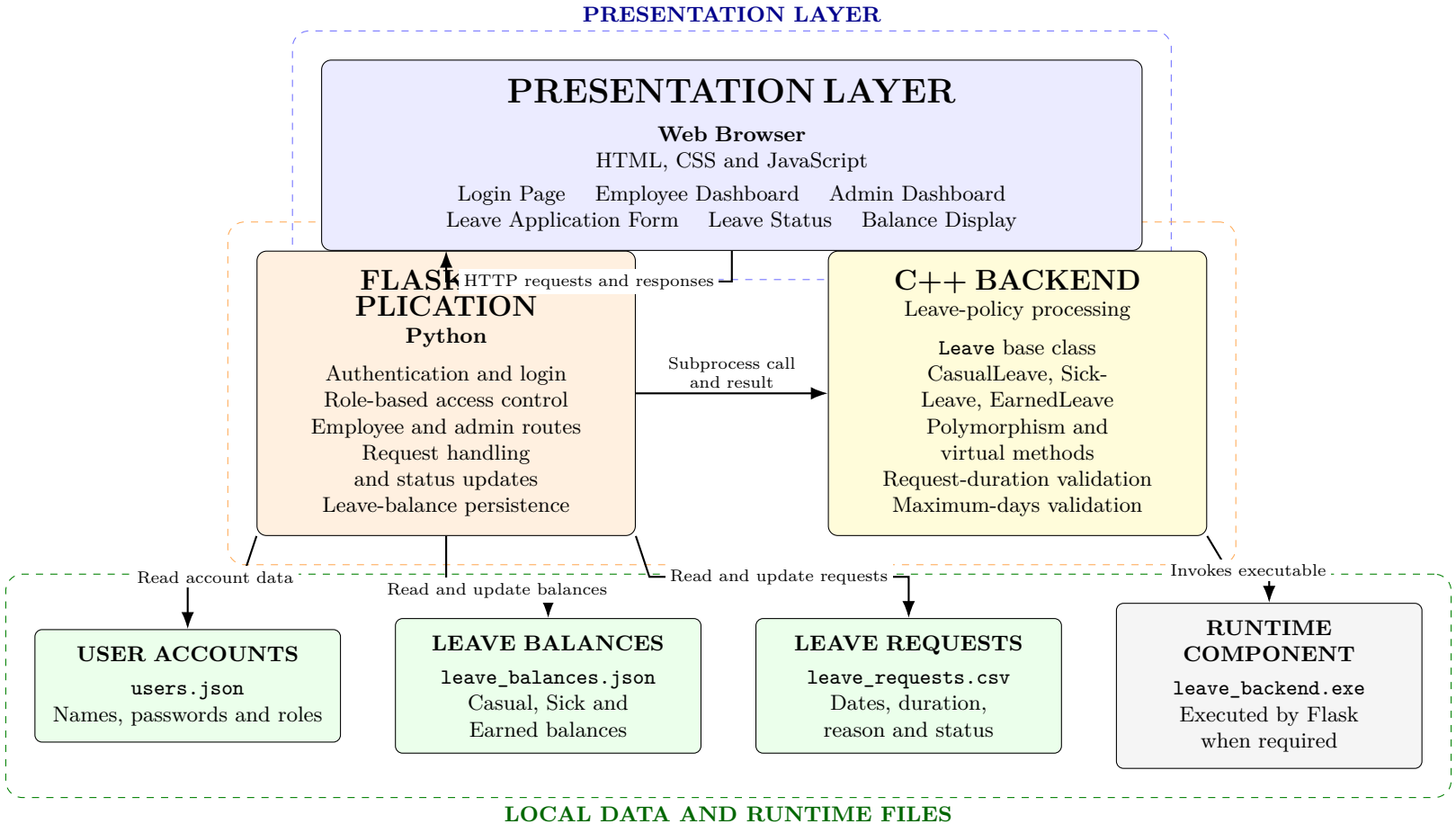

PRESENTATION LAYER
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Web Browser
HTML, CSS and JavaScript
Login Page
Employee Dashboard
Admin Dashboard
Leave Application Form
Leave Status
Balance Display
HTTP requests and responses
FLASK
PLICATION
Python
Authentication and login
Role-based access control
Employee and admin routes
Request handling
and status updates
Leave-balance persistence
Subprocess call
and result
C++ BACKEND
Leave-policy processing
Leave base class
CasualLeave, Sick-
Leave, EarnedLeave
Polymorphism and
virtual methods
Request-duration validation
Maximum-days validation
Read account data
Read and update balances
Read and update requests
Invokes executable
USER ACCOUNTS
users.json
Names, passwords and roles
LEAVE BALANCES
leave_balances.json
Casual, Sick and
Earned balances
LEAVE REQUESTS
leave_requests.csv
Dates, duration,
reason and status
RUNTIME
COMPONENT
leave_backend.exe
Executed by Flask
when required
LOCAL DATA AND RUNTIME FILES